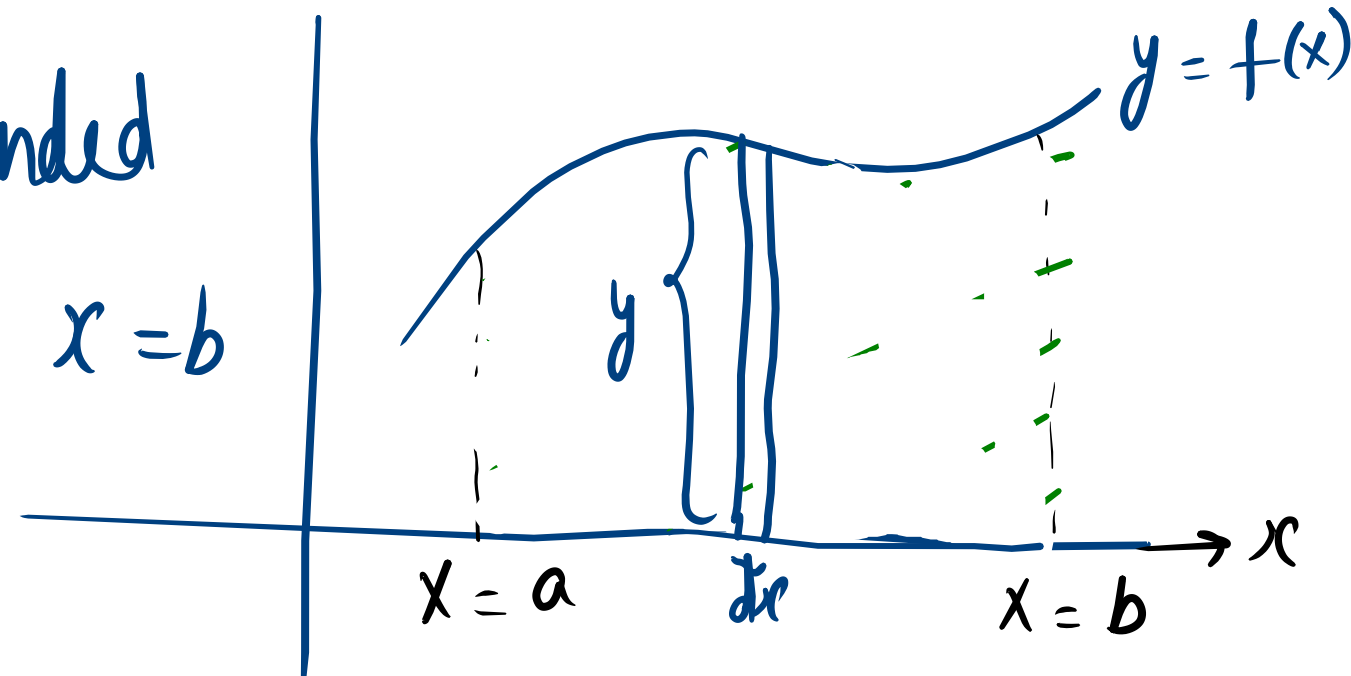


I

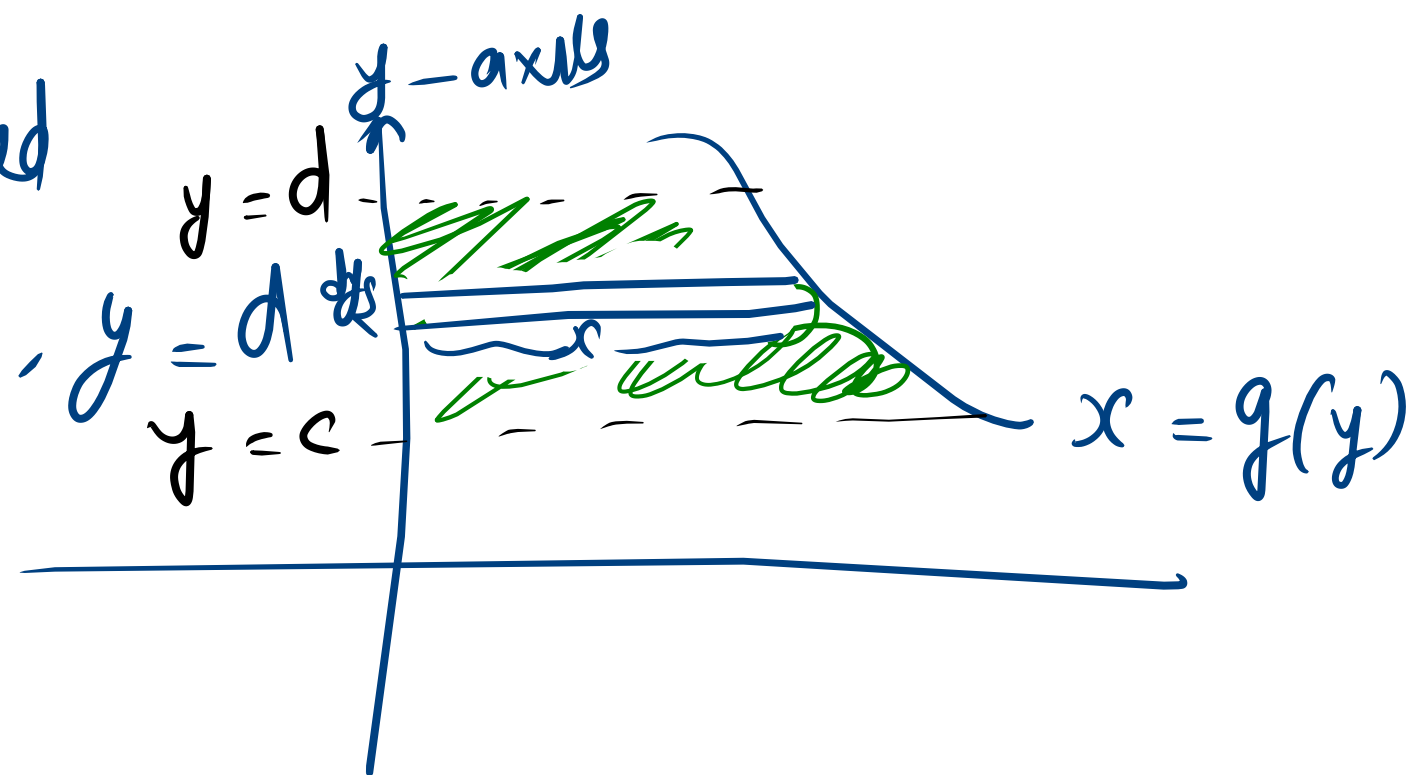
A) $\int_a^b f(x) dx$
or
 $\int_a^b y dx$

= Area of the region bounded
by $y = f(x)$, $x = a$, $x = b$
and x -axis



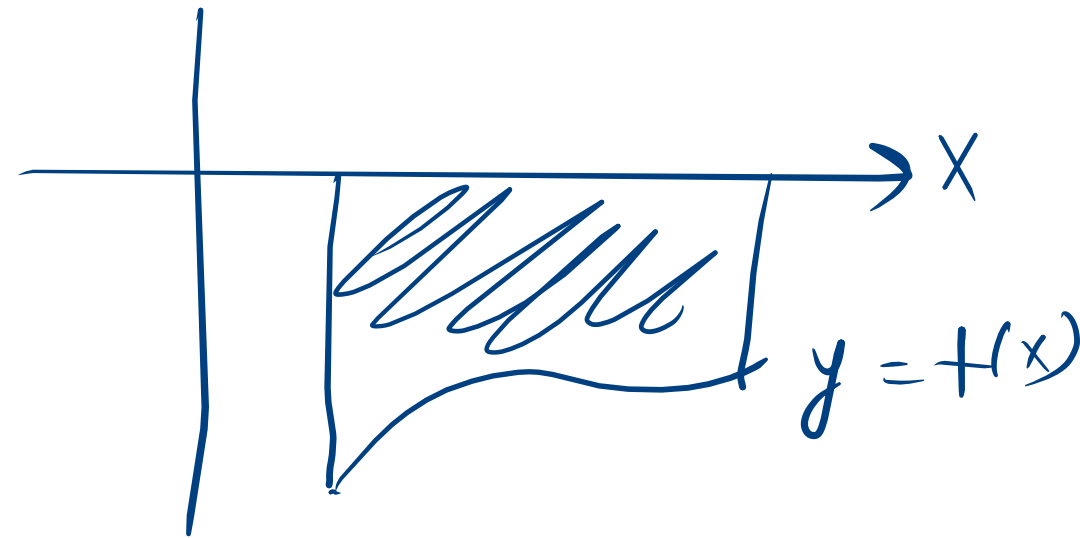
B) $\int_c^d g(y) dy$
or
 $\int_c^d x dy$

= Area of region bounded
by $x = g(y)$, $y = c$, $y = d$
and y -axis



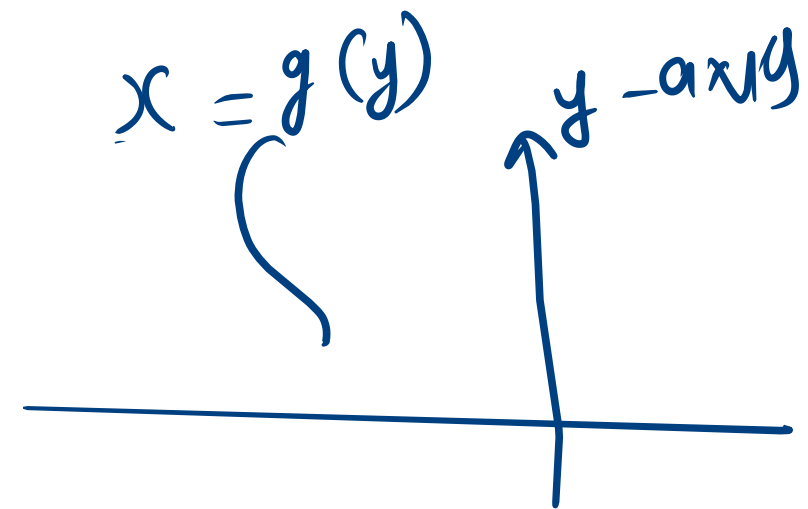
10 Area = $\left| \int_a^b f(x) dx \right|$ or $\left| \int_a^b y dx \right|$

only if $y = f(x)$ is below x-axis

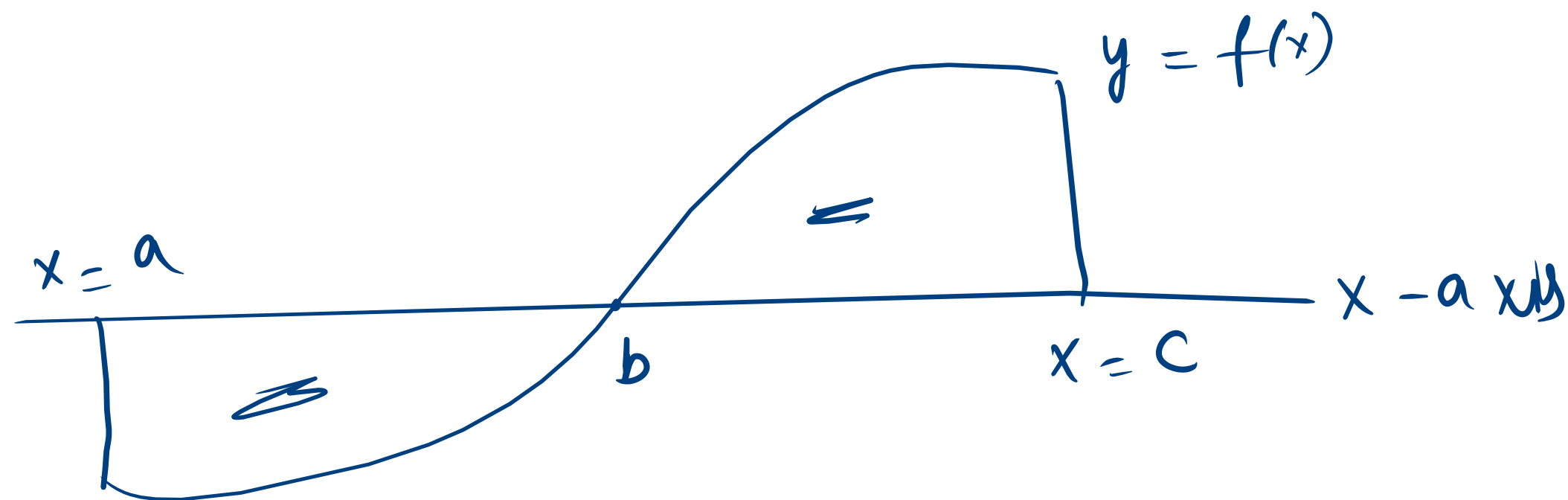


11 Area = $\left| \int_c^d g(y) dy \right|$ or $\left| \int_c^d x dy \right|$

if $x = g(y)$ is on left side of y-axis



Ex



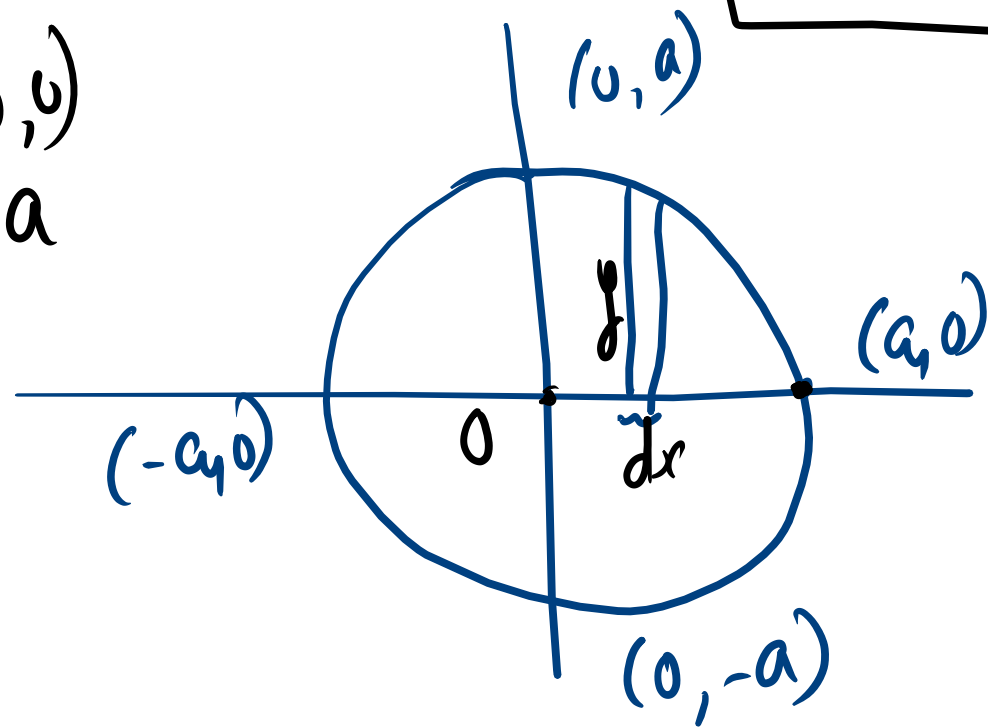
$$\text{Area} = \left| \int_a^b f(x) dx \right| + \int_b^c f(x) dx$$

↙
below x-axis

Example 1 Find the area enclosed by the circle $x^2 + y^2 = a^2$.

Step I Draw Curve

Centre = $(0, 0)$
radius = a



Step II

$$\begin{aligned}\text{Area} &= 4 \times \text{Area of region in I quad.} \\ &= 4 \int_0^a y \, dx\end{aligned}$$

$$\begin{aligned}x^2 + y^2 &= r^2 \\ \downarrow \\ \text{Centre} &= (0, 0) \\ \text{radius} &= r\end{aligned}$$

$$x^2 + y^2 = a^2$$

$$y^2 = a^2 - x^2$$

$$y = \pm \sqrt{a^2 - x^2}$$

I quad, $y \rightarrow +ve$

$$y = \sqrt{a^2 - x^2}$$

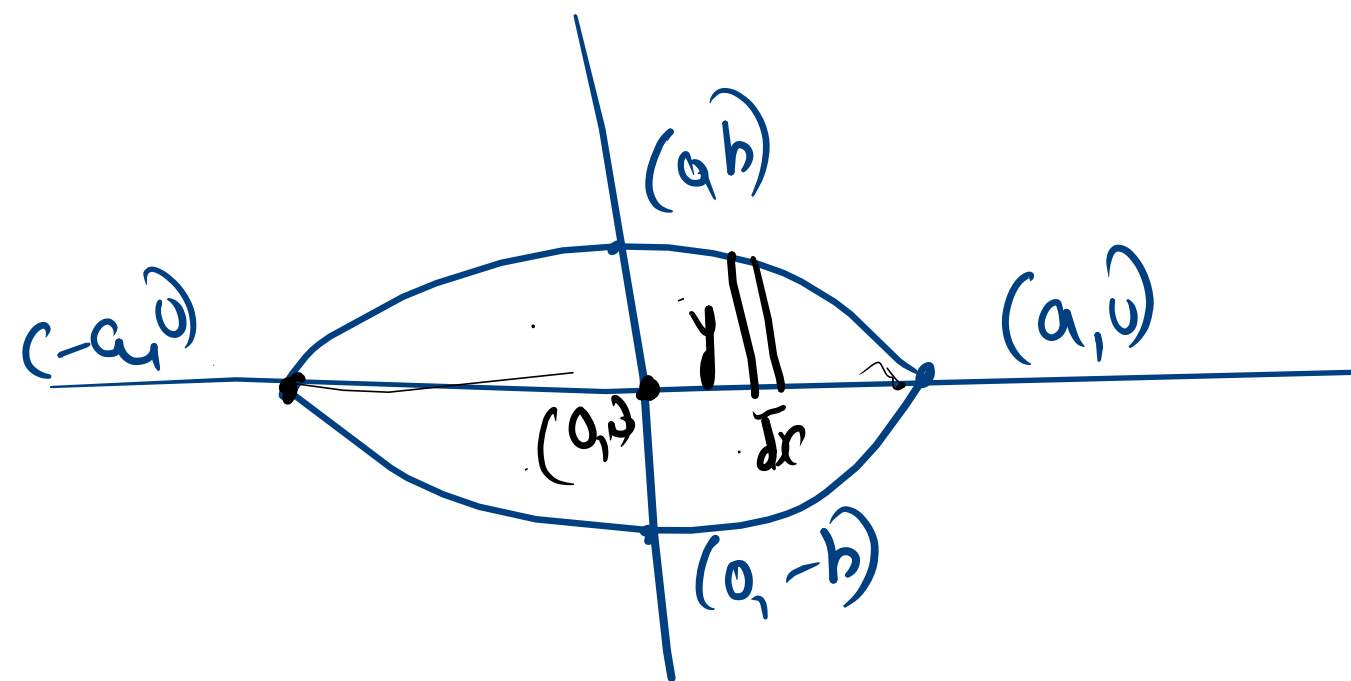
$$\text{Area} = 4 \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$= 4 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

$$= 4 \left[0 + \frac{a^2}{2} \cdot \frac{\pi}{2} - 0 - 0 \right]$$

$$\text{Area} = \pi a^2$$

Example 2 Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ Fig 8



$$\begin{aligned} \text{Area} &= 4 \times \text{Area of I quadrant} \\ &= 4 \int_0^a y \, dx \end{aligned}$$

$$\begin{aligned} \frac{y^2}{b^2} &= 1 - \frac{x^2}{a^2} \\ \frac{y^2}{b^2} &= \frac{a^2 - x^2}{a^2} \end{aligned}$$

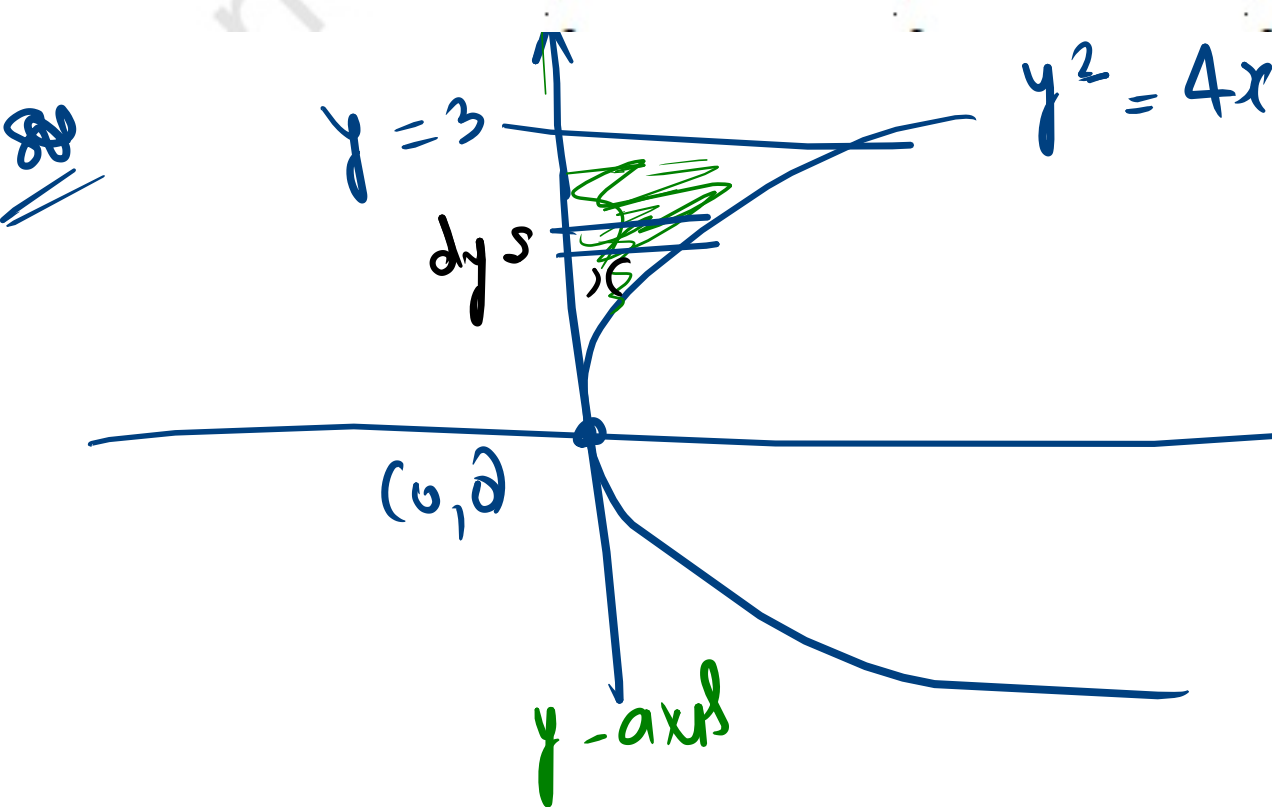
$$y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

I quad, $y > 0$

$$\text{Area} = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

4. Area of the region bounded by the curve $y^2 = 4x$, y -axis and the line $y = 3$ is



$$\text{Area} = \int_0^3 x \, dy$$

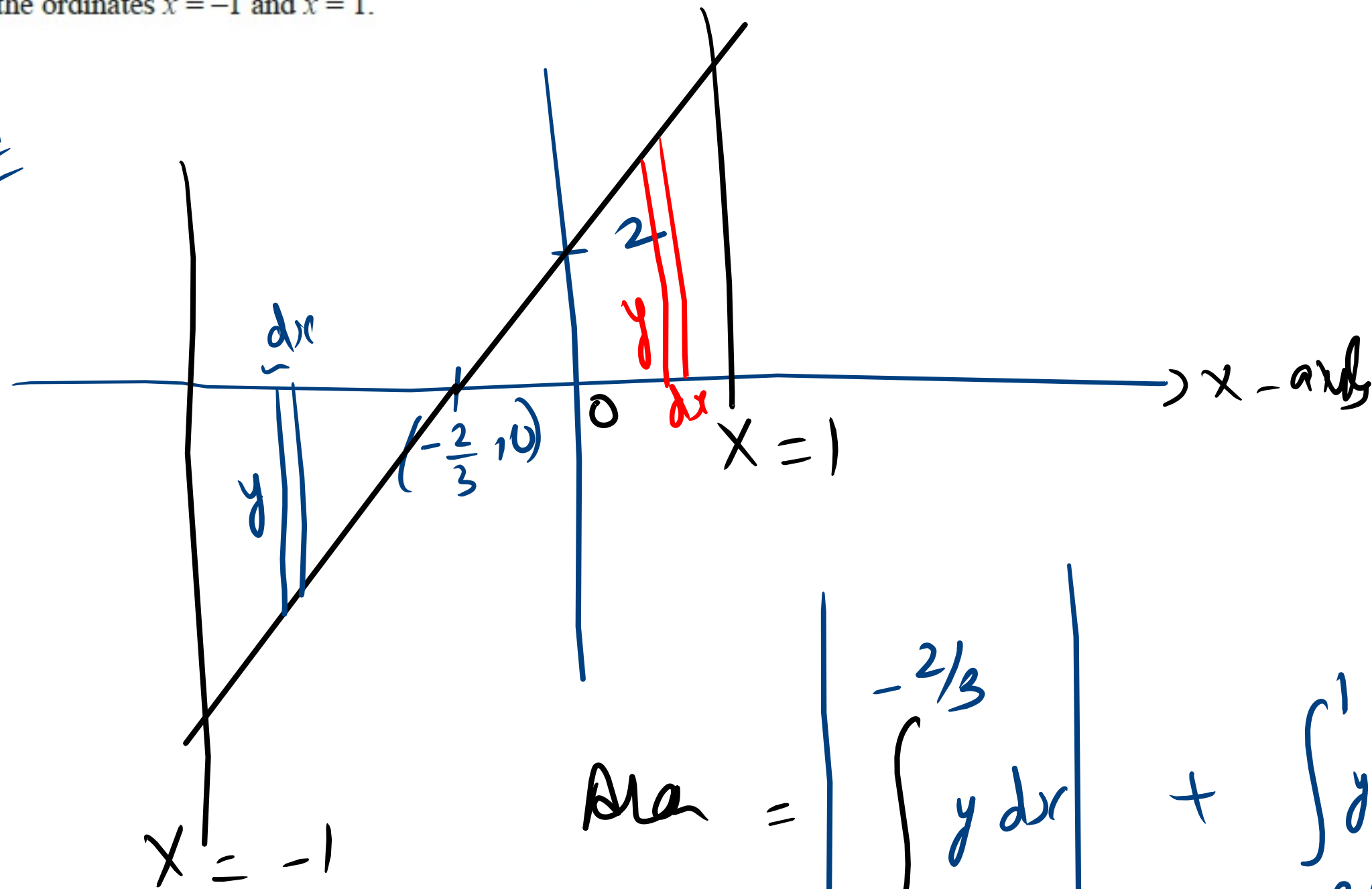
$$y^2 = 4x$$

$$x = \frac{y^2}{4}$$

$$\text{Area} = \int_0^3 \frac{y^2}{4} \, dy$$

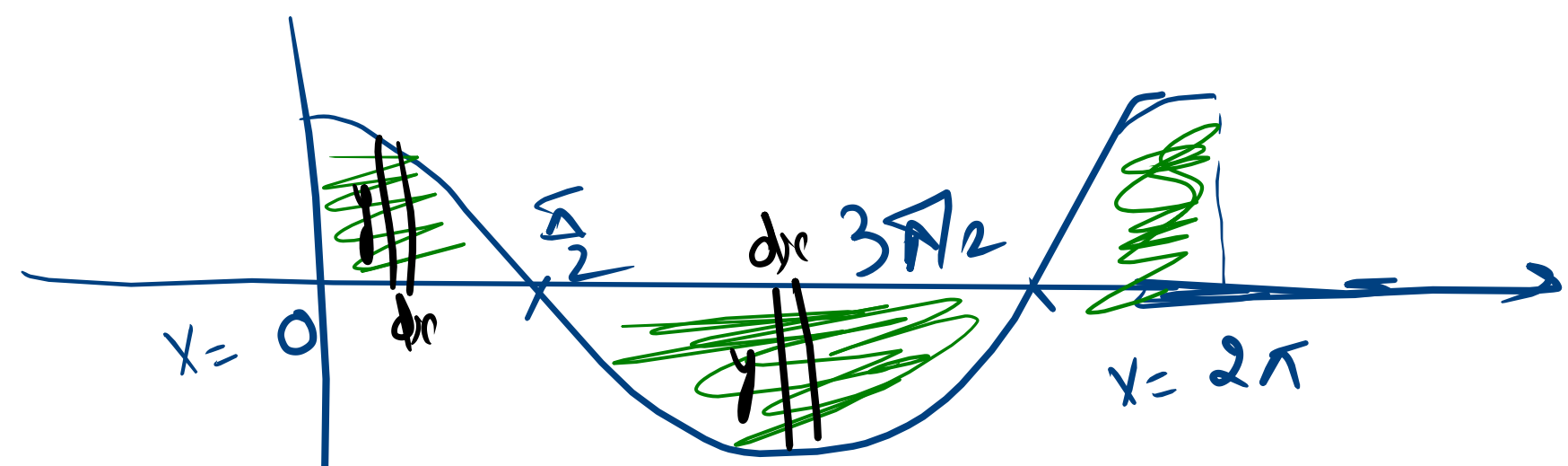
Example 3 Find the area of the region bounded by the line $y = 3x + 2$, the x -axis and the ordinates $x = -1$ and $x = 1$.

$$y = 3x + 2$$



$$\text{Area} = \left| \int_{-1}^{-2/3} y \, dx \right| + \int_{-2/3}^1 y \, dx$$

Example 4 Find the area bounded by the curve $y = \cos x$ between $x = 0$ and $x = 2\pi$.



$$\text{Area} = \int_0^{\pi/2} y \, dx + \left| \int_{\pi/2}^{3\pi/2} y \, dx \right| + \int_{3\pi/2}^{2\pi} y \, dx$$

but $y = \cos x$

$$\int_0^{\pi/2} \cos x \, dx + \left| \int_{\pi/2}^{3\pi/2} \cos x \, dx \right| + \int_{3\pi/2}^{2\pi} \cos x \, dx$$

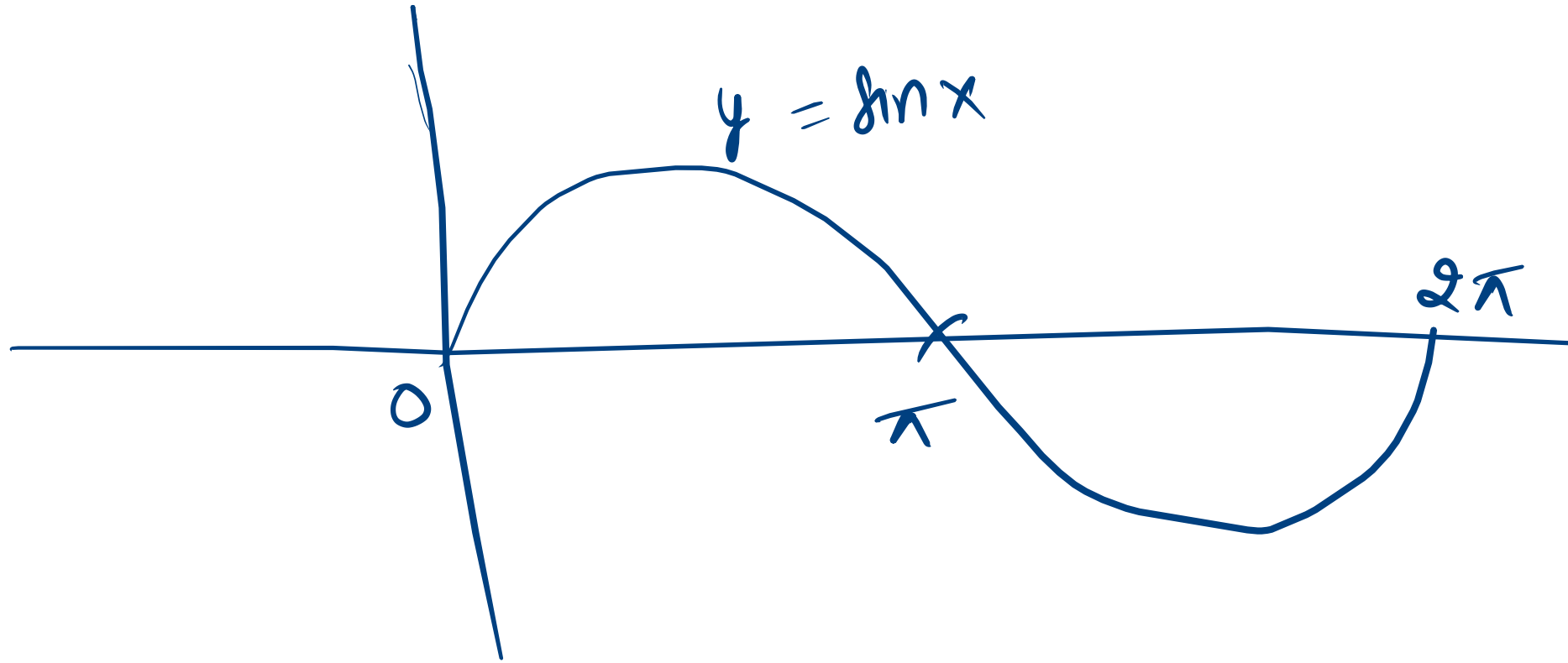
$$\left[\sin x \right]_0^{\pi/2} + \left| \left[\sin x \right]_{\pi/2}^{3\pi/2} \right| + \left[\sin x \right]_{3\pi/2}^{2\pi}$$

$$\left| \sin \frac{3\pi}{2} - \sin \frac{\pi}{2} \right|$$

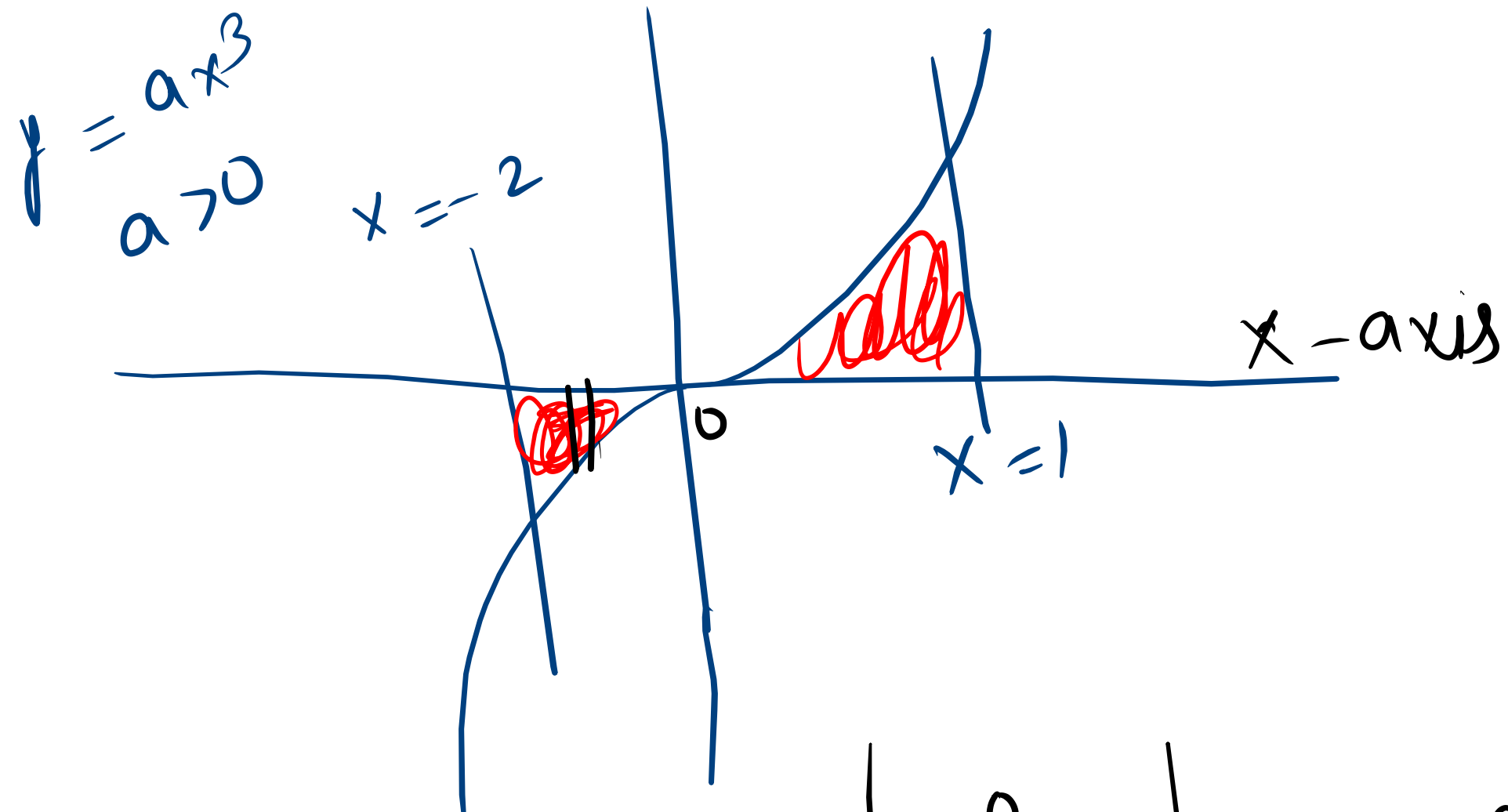
$$|-1 - 1|$$

$$2$$

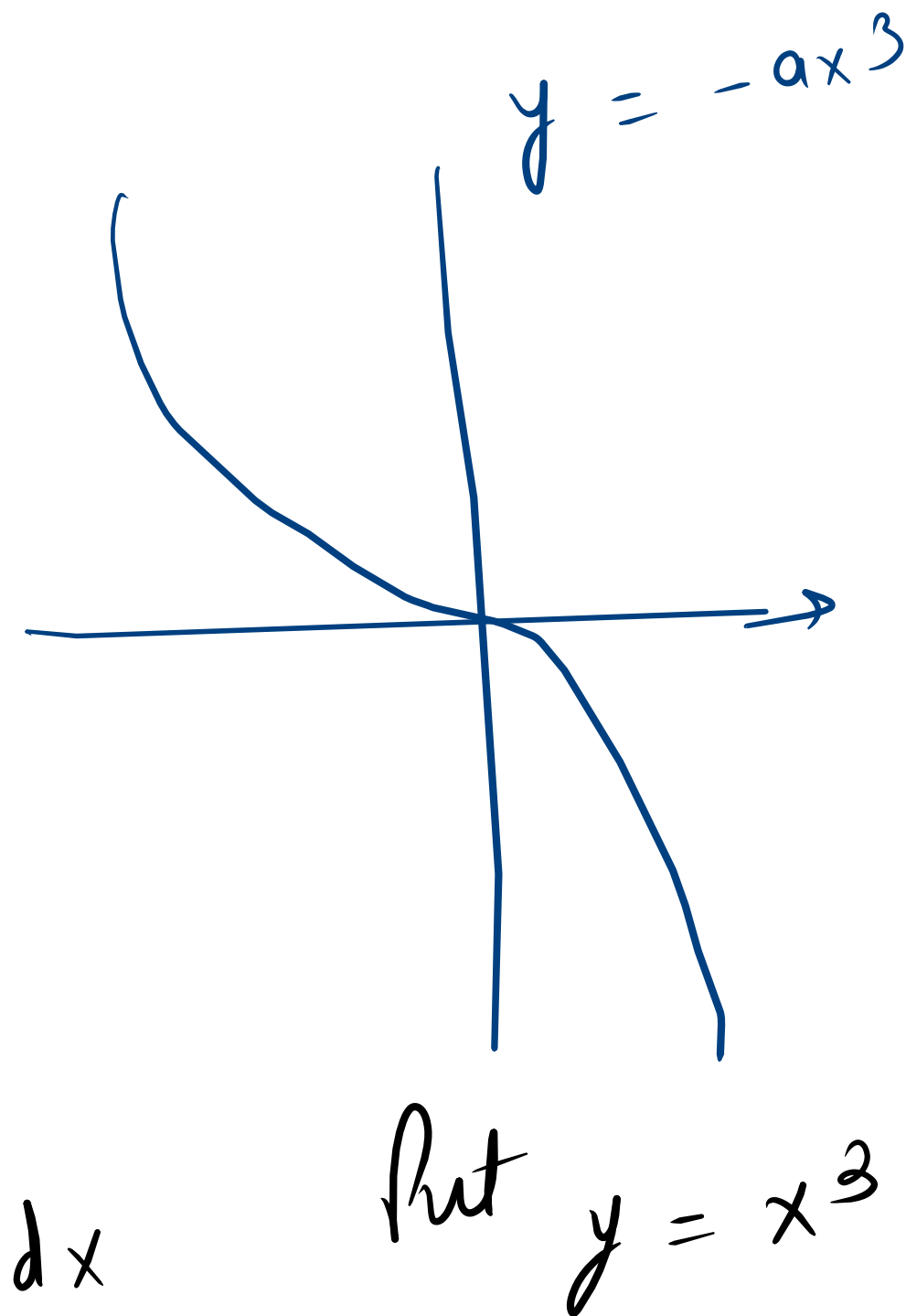
3. Find the area bounded by the curve $y = \sin x$ between $x = 0$ and $x = 2\pi$.



4. Area bounded by the curve $y = x^3$, the x -axis and the ordinates $x = -2$ and $x = 1$ is

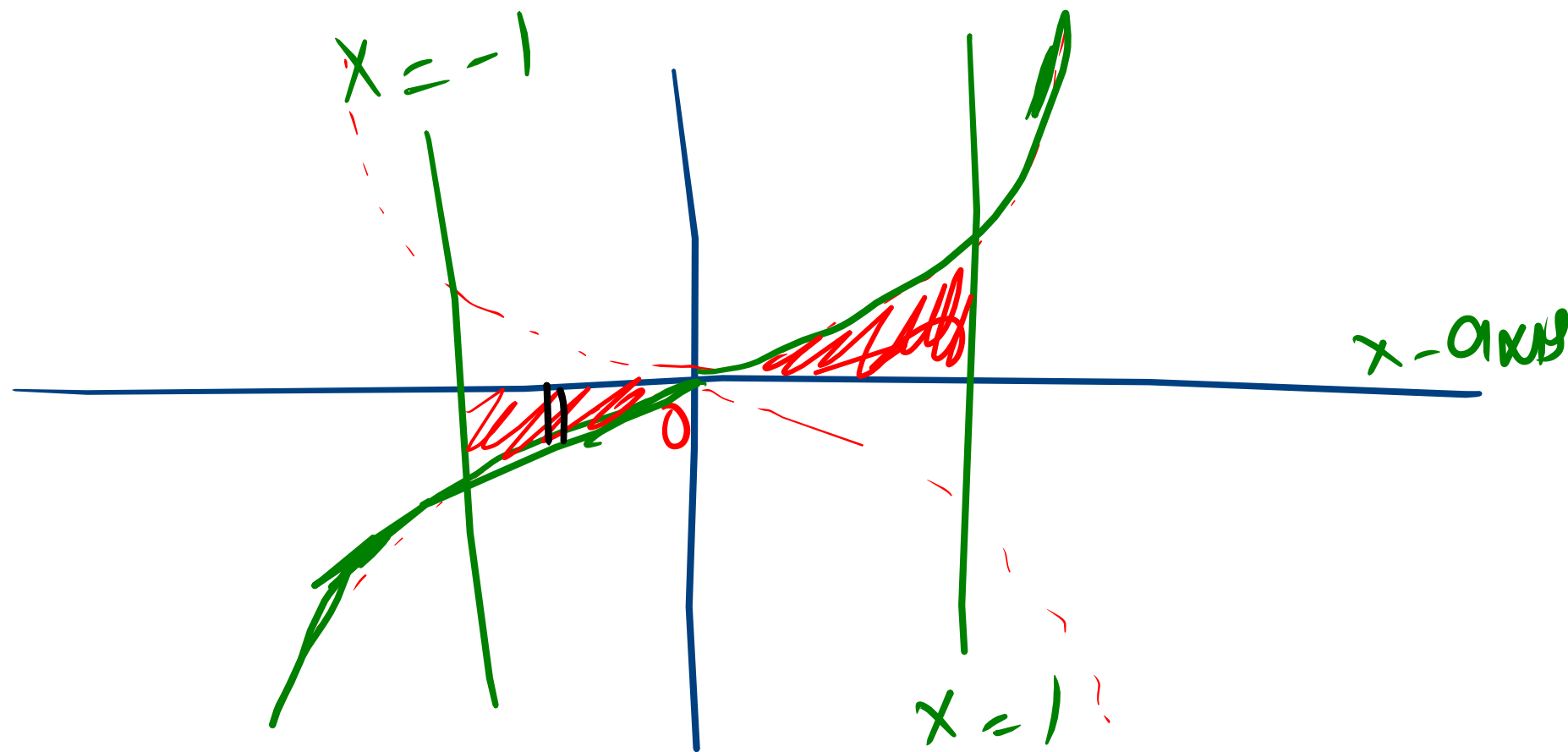


$$\text{Area} = \left| \int_{-2}^0 y \, dx \right| + \int_0^1 y \, dx$$



5. The area bounded by the curve $y = x|x|$, x -axis and the ordinates $x = -1$ and $x = 1$ is given by

$$y = x|x| = \begin{cases} x(x) & x \geq 0 \\ x(-x) & x < 0 \end{cases} = \begin{cases} x^2 & ; x \geq 0 \\ -x^2 & ; x < 0 \end{cases}$$



$$\text{Area} = \left| \int_{-1}^0 y \, dx \right| + \int_0^1 y \, dx$$

$$\text{Area} = \left| \int_{-1}^0 (-x^2) \, dx \right| + \int_0^1 x^2 \, dx$$

Homework

Chapter-8

↳ Example - 1, 2, 3, 4

Exercise 8-1 → 01 to 04

Misc. Exor → 01 to 05

for sat

Chapter 7

Example - 41, 42

Misc exercise - 029 to 040

Exercise - 7.5, 7.6